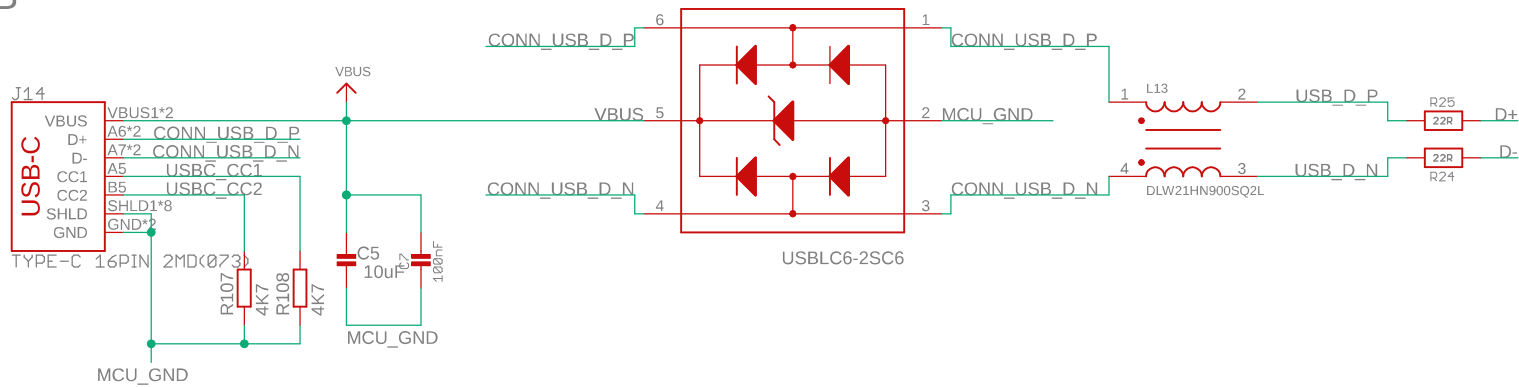
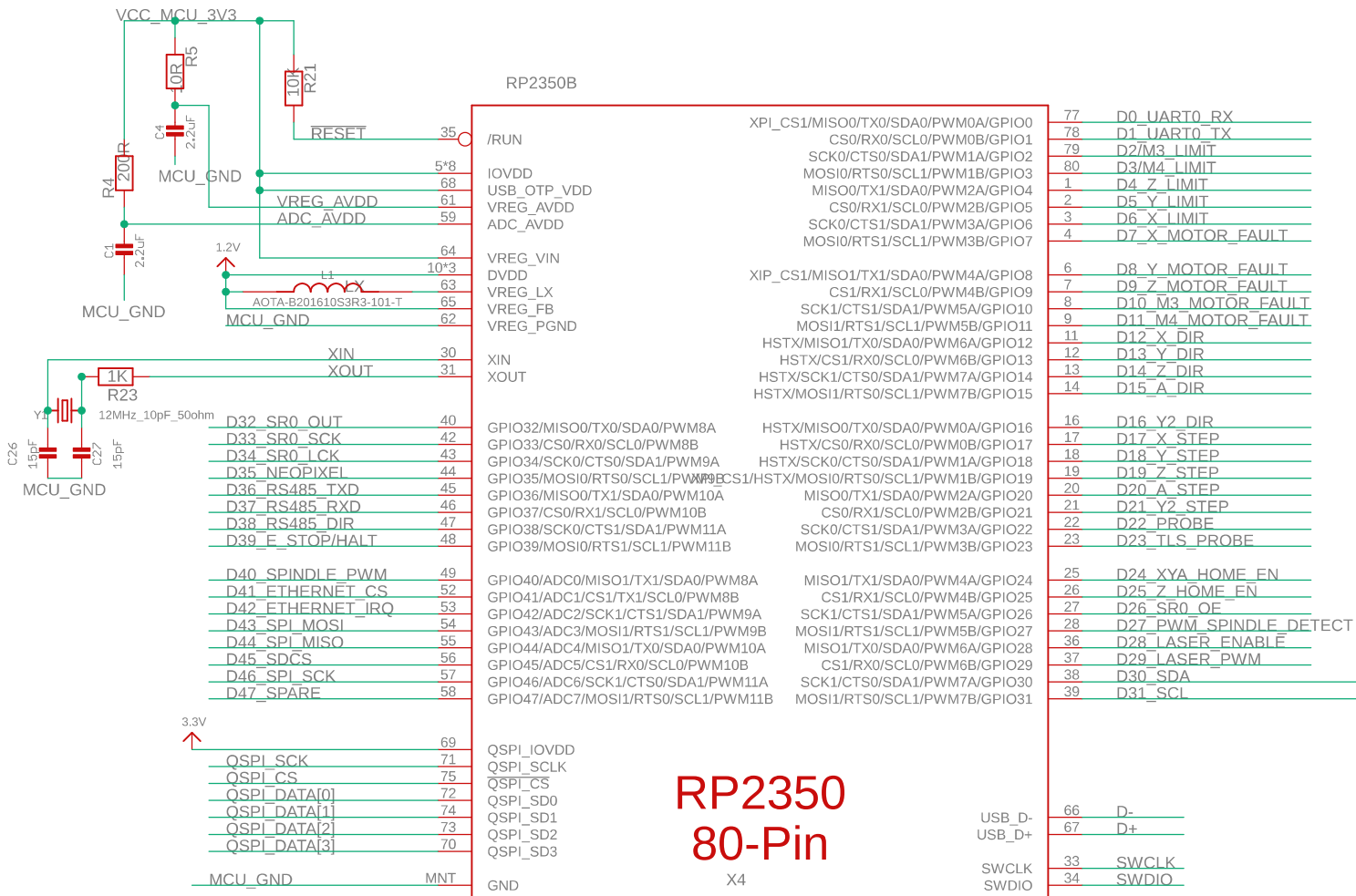
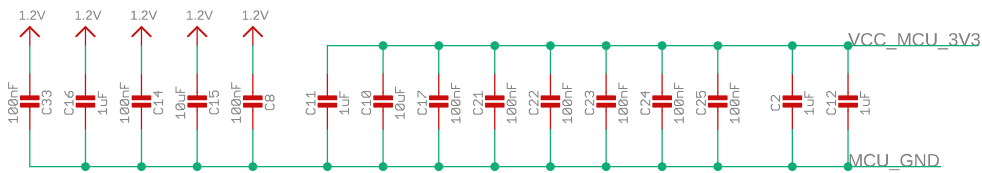


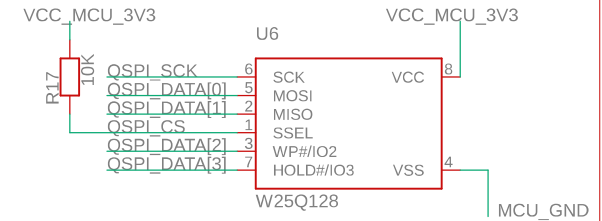
USB



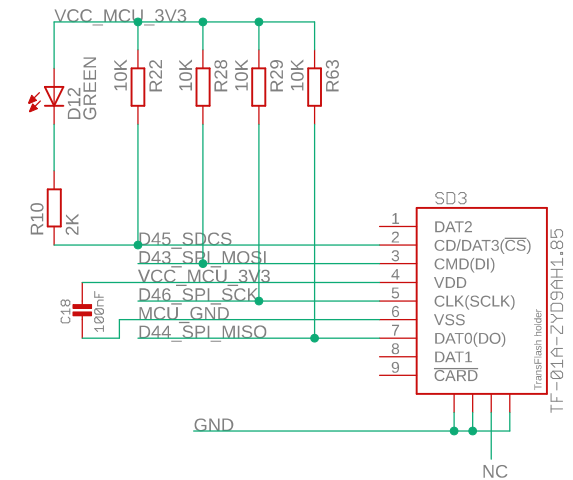
MCU



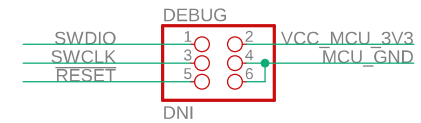
EEPROM



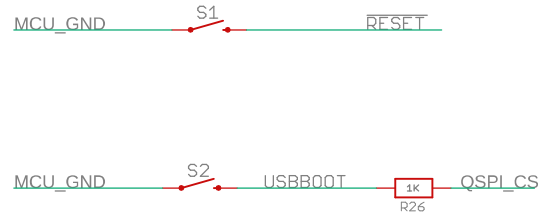
SD CARD



DEBUG

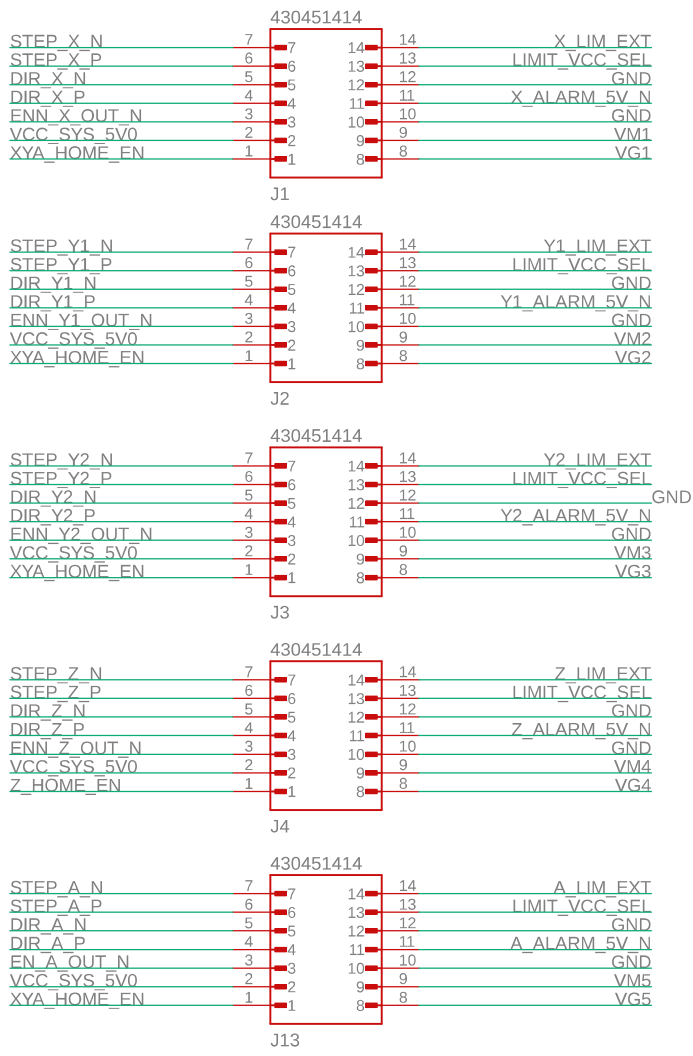
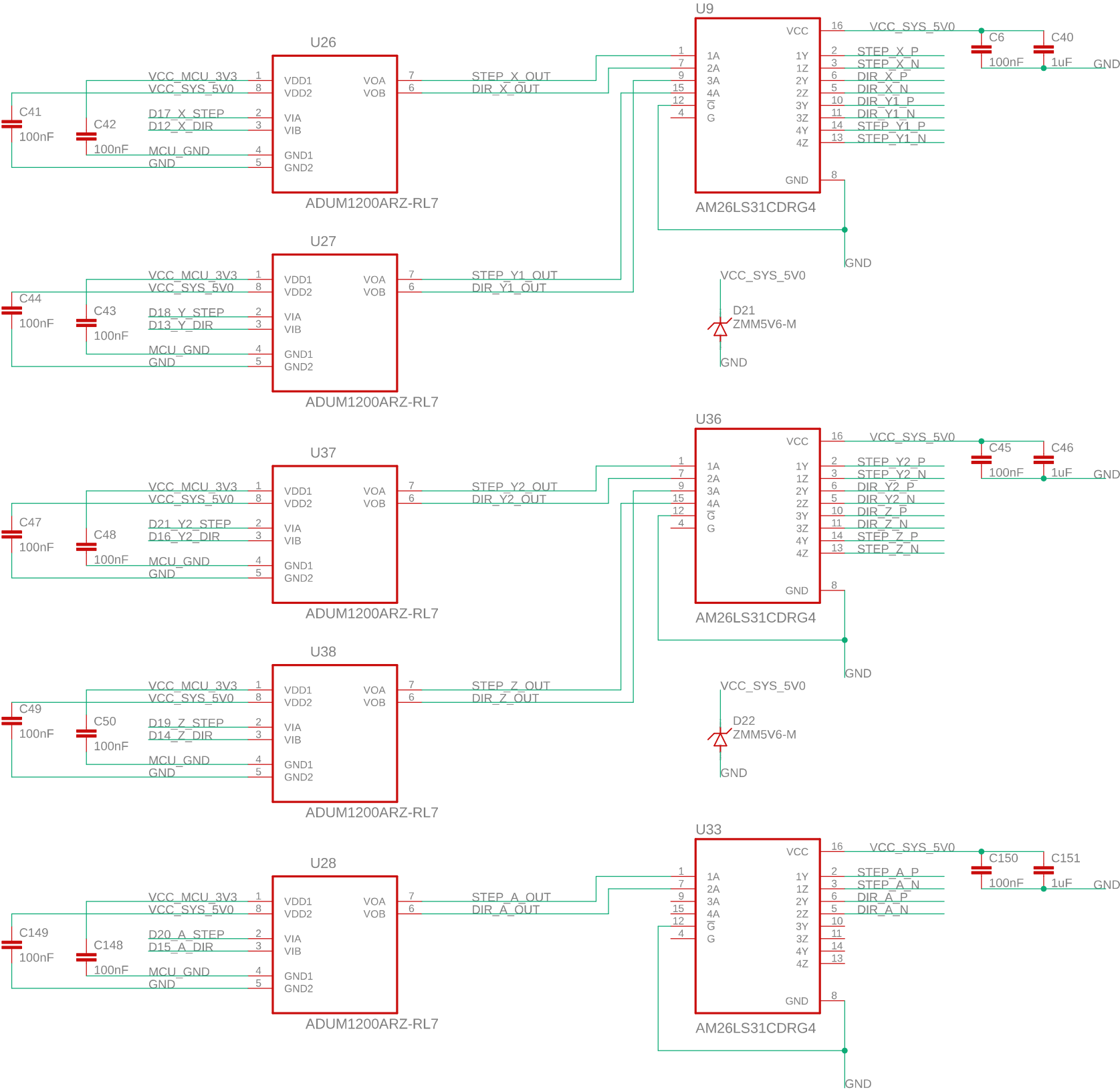


DFU/RESET



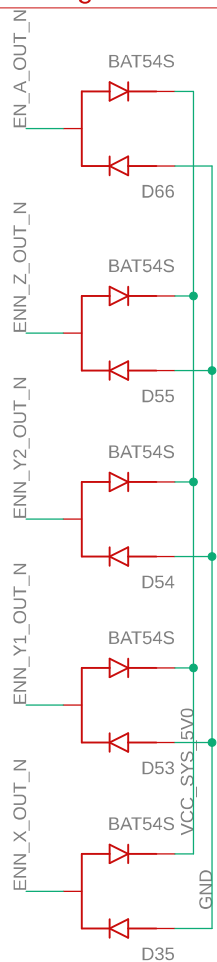
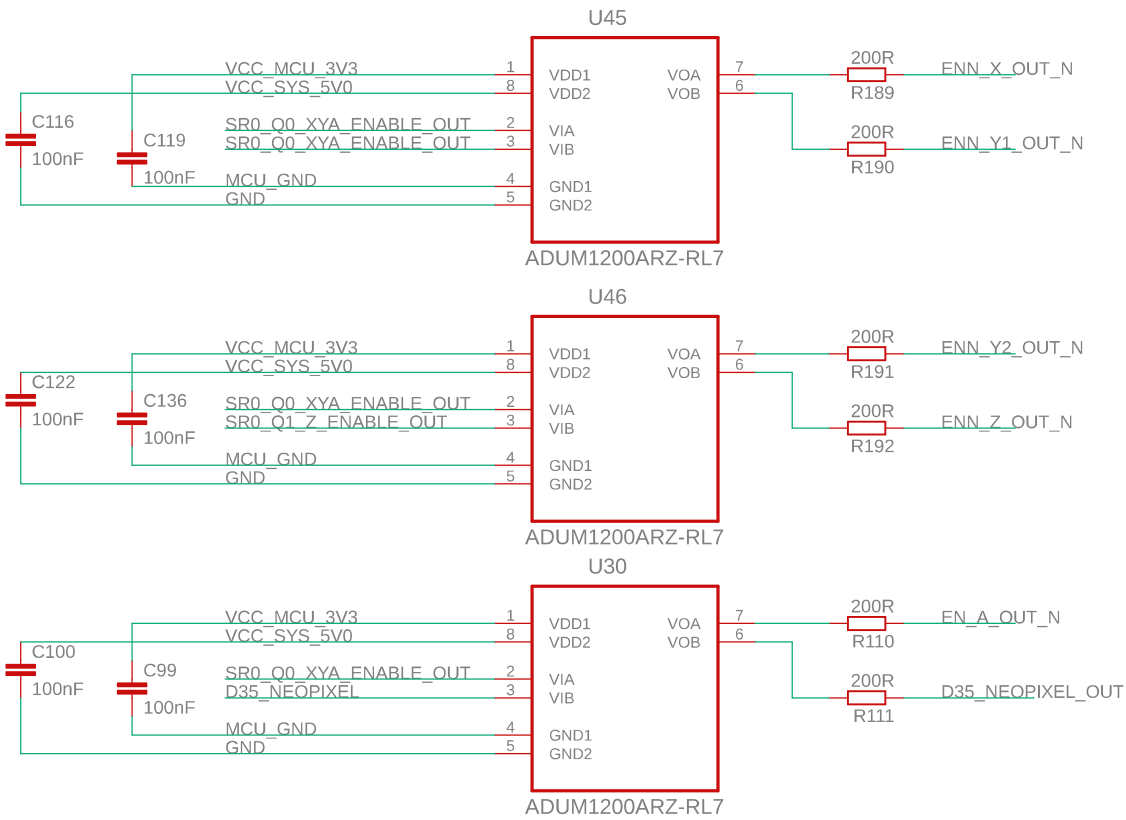
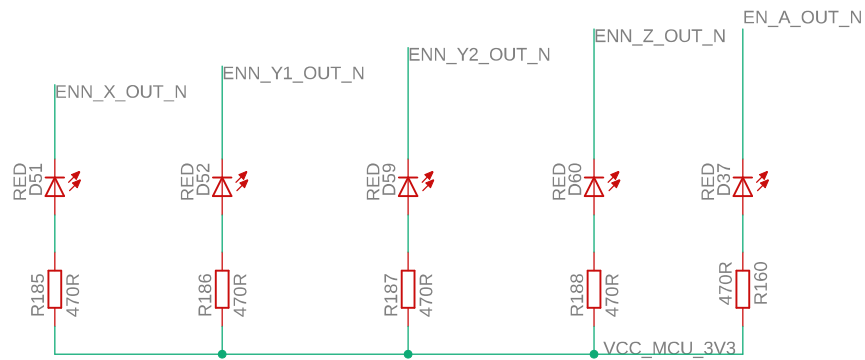
slb-ext - proto3.1 - SLB-Lite Version - Eagle V7 version	
2026/09/21 15:42	
Sheet: 1/12	

MOTOR STEP DIR BUFFER

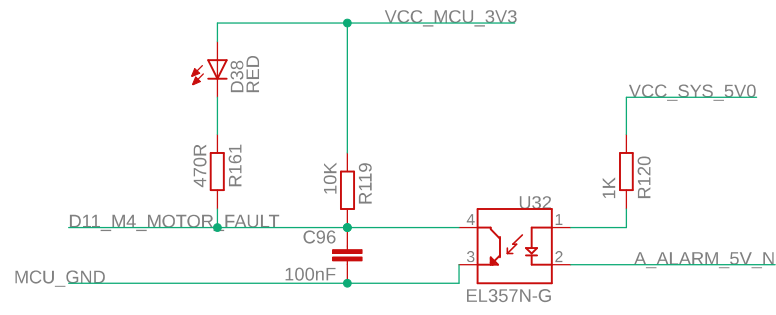
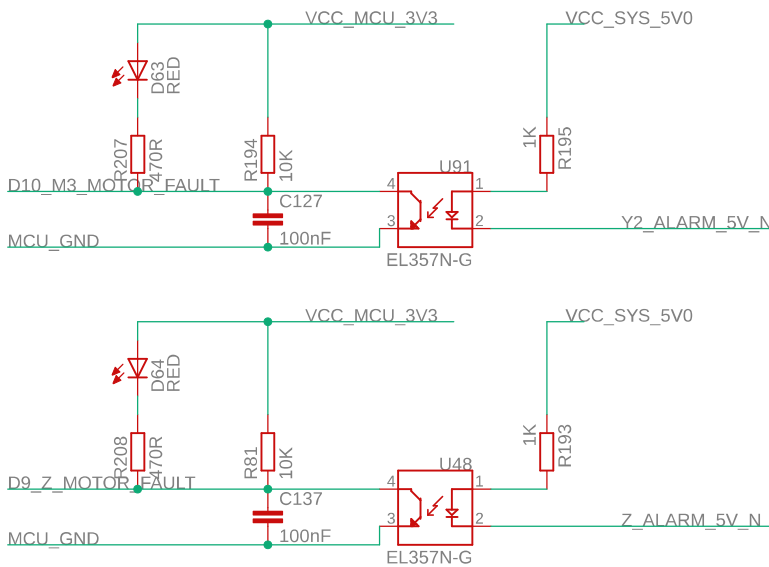
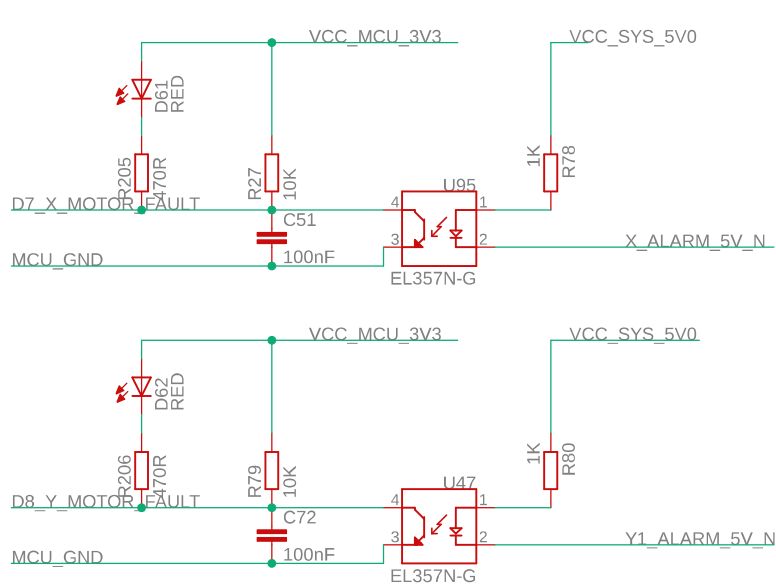


MOTOR ENABLE BUFFER

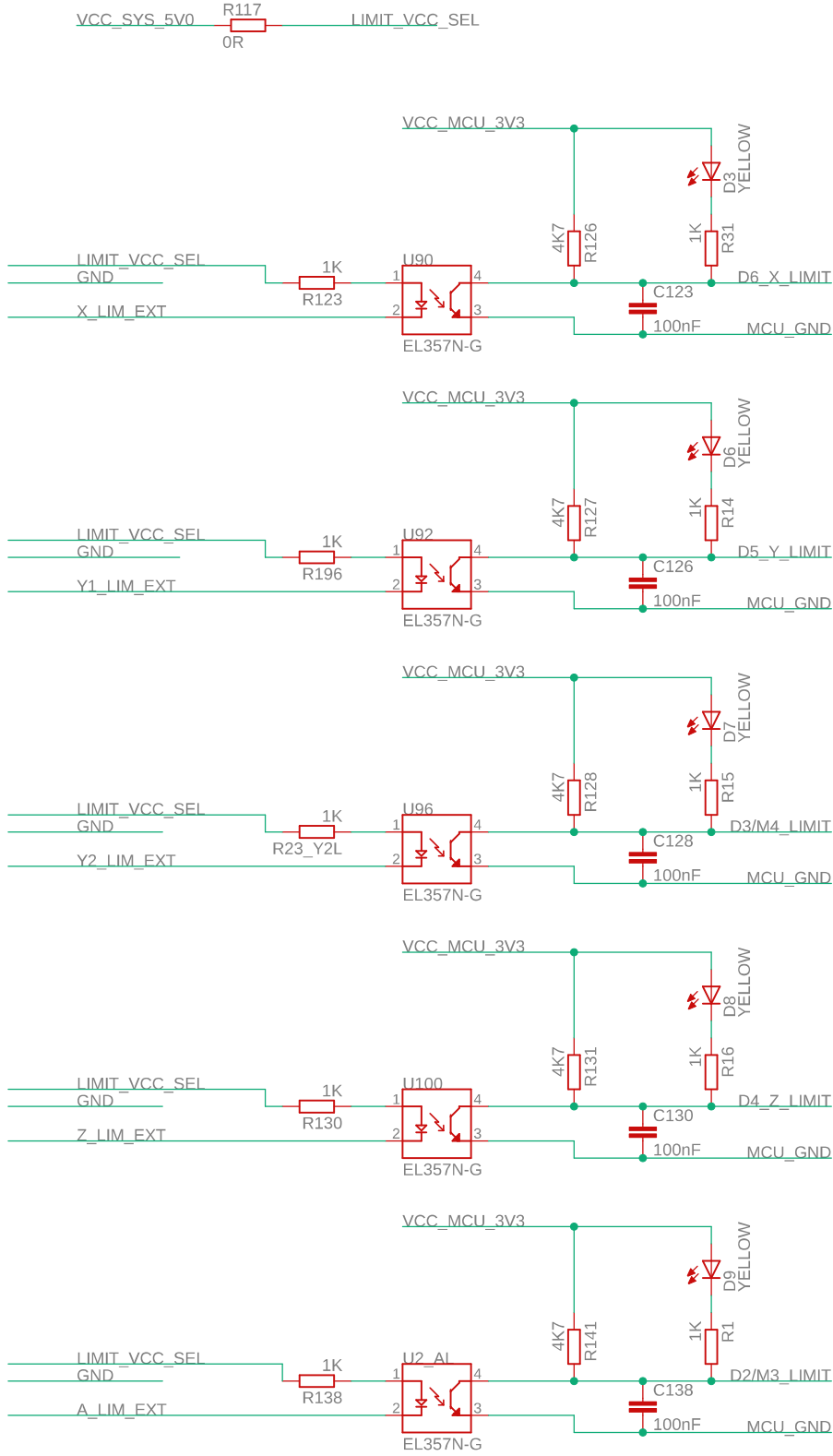
NOTE: SIENCI CLS MOTORS ARE ENABLE BY DEFAULT
APPLYING VOLTAGE BETWEEN EN+AND EN- "DISABLES" THE MOTOR



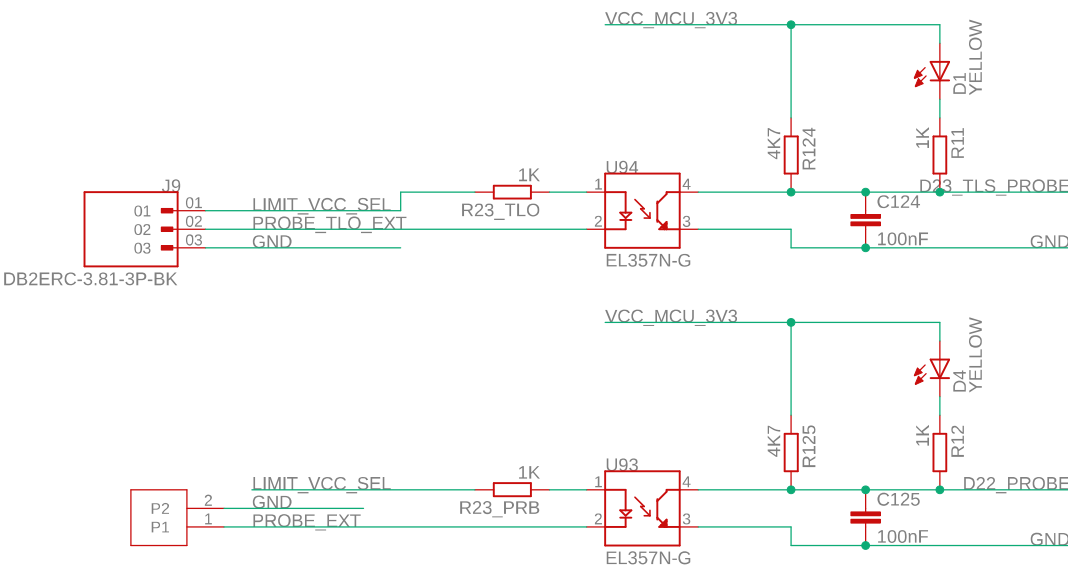
MOTOR ALARM BUFFER



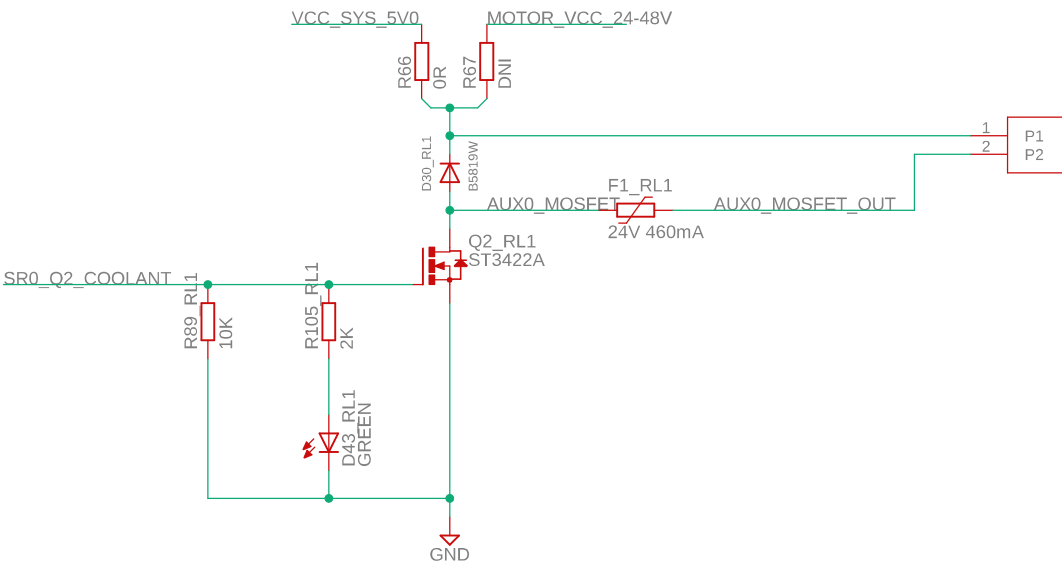
LIMIT SWITCHES



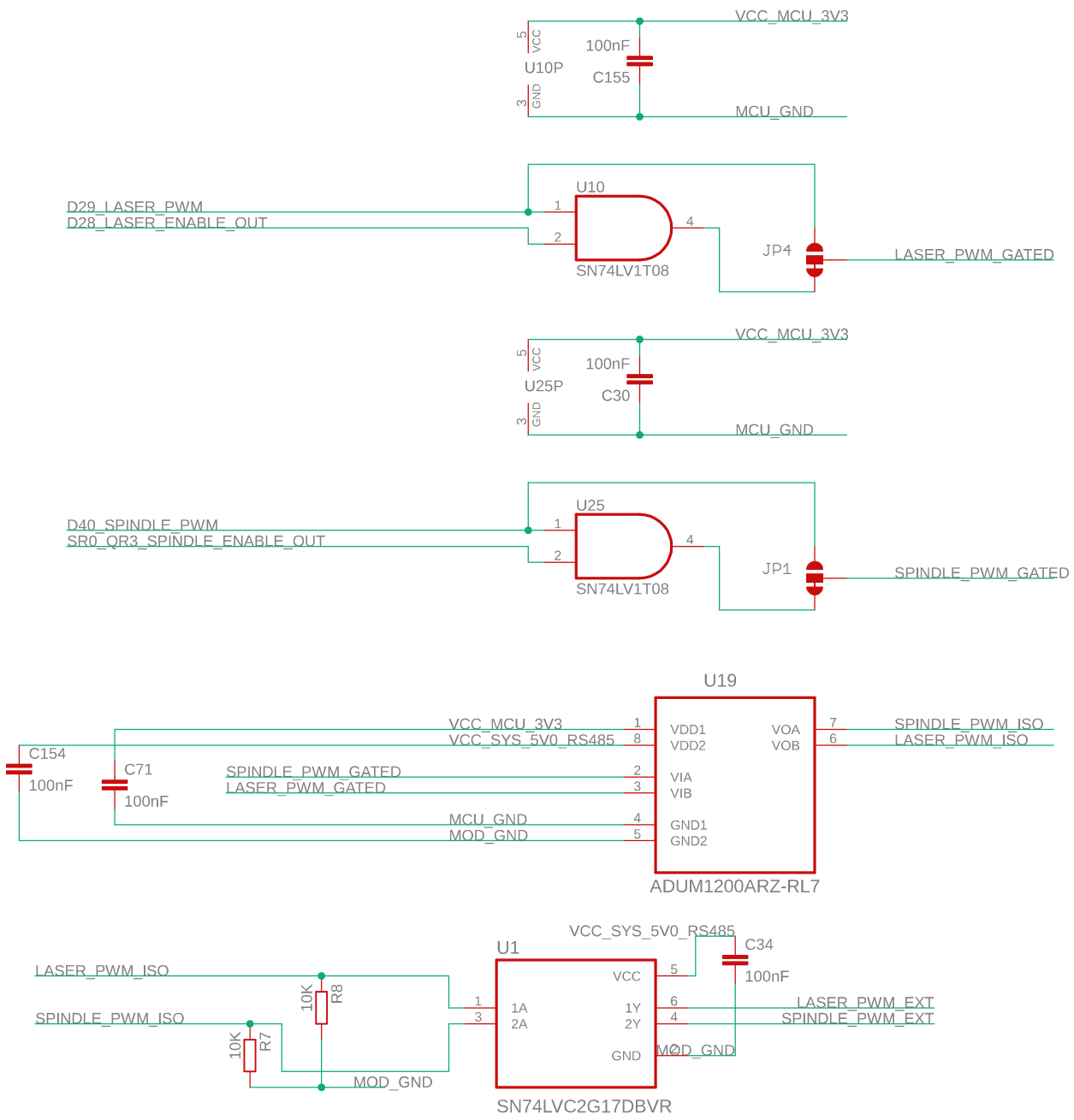
PROBES



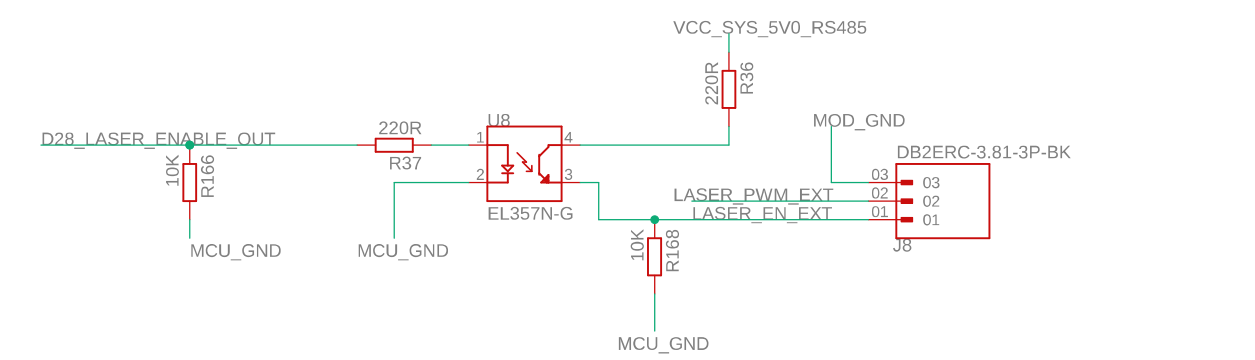
MOSFETS



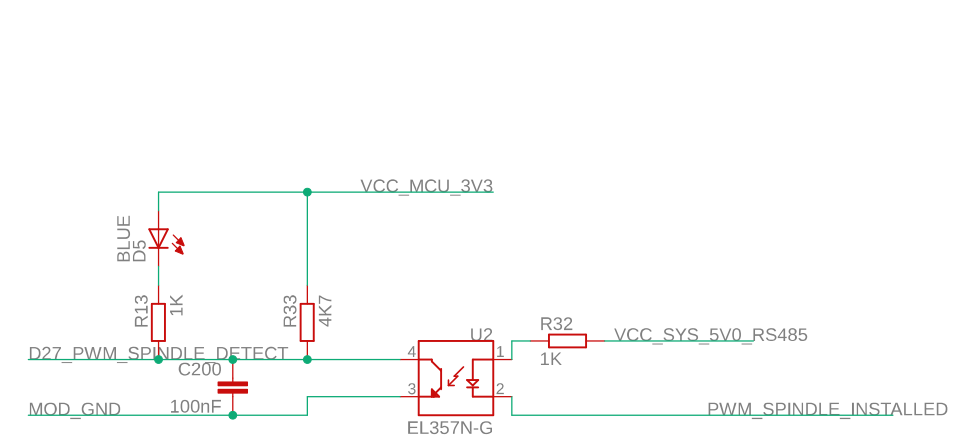
LASER/SPINDLE SAFETY GATING AND ISOLATION



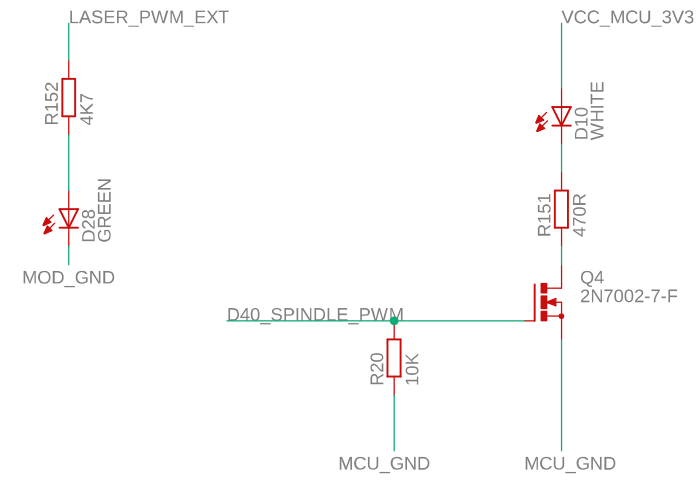
LASER



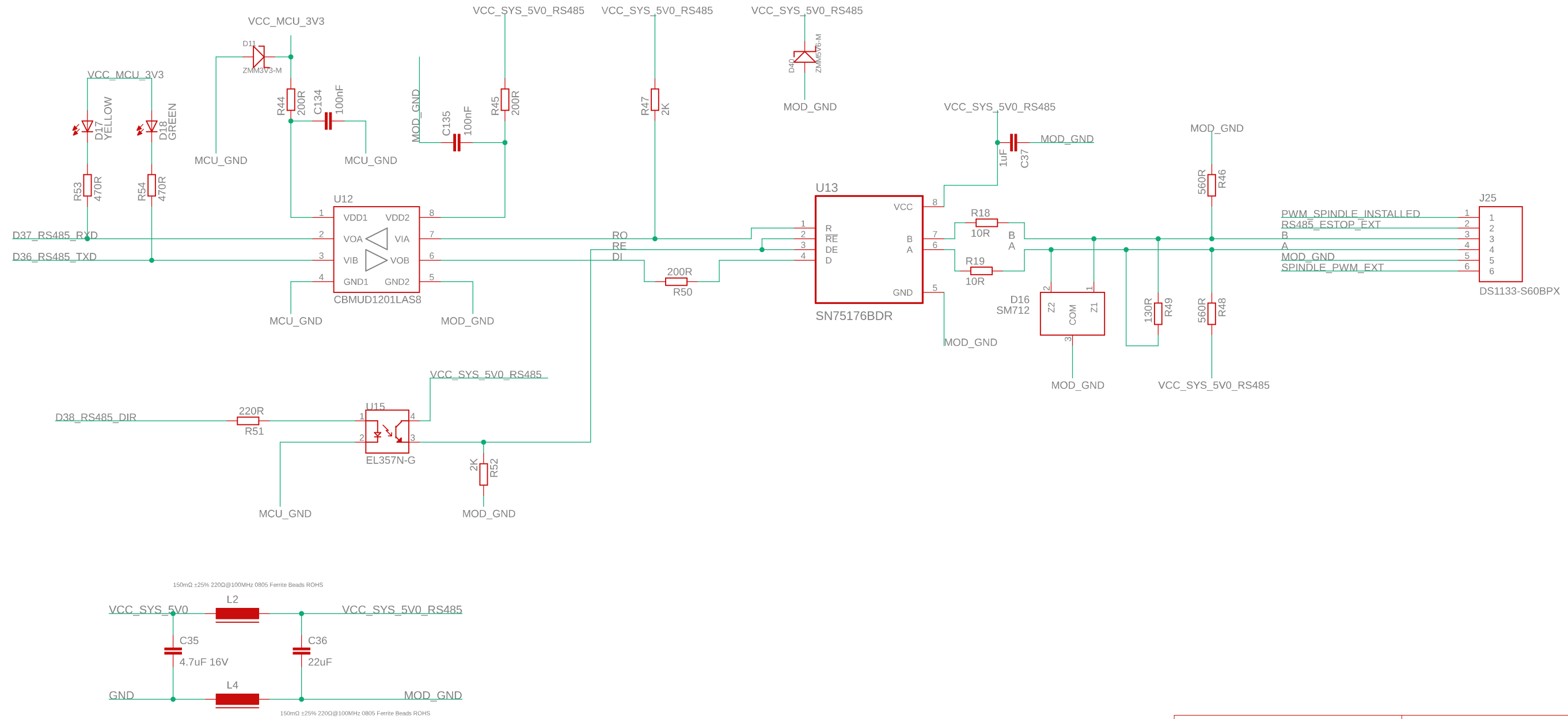
PWM Spindle Detect



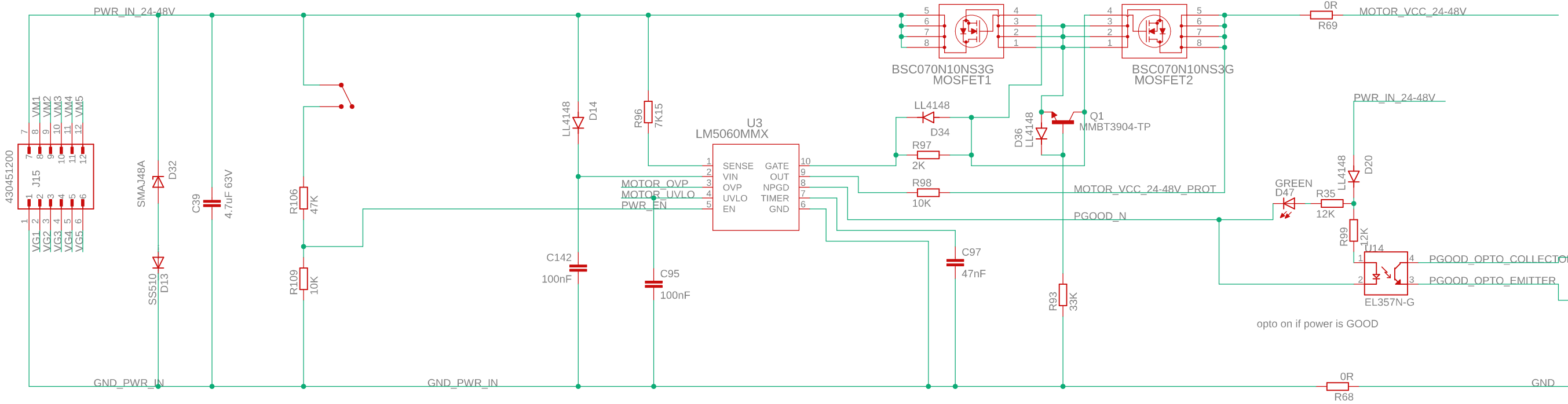
LASER/SPINDLE INDICATOR LEDS



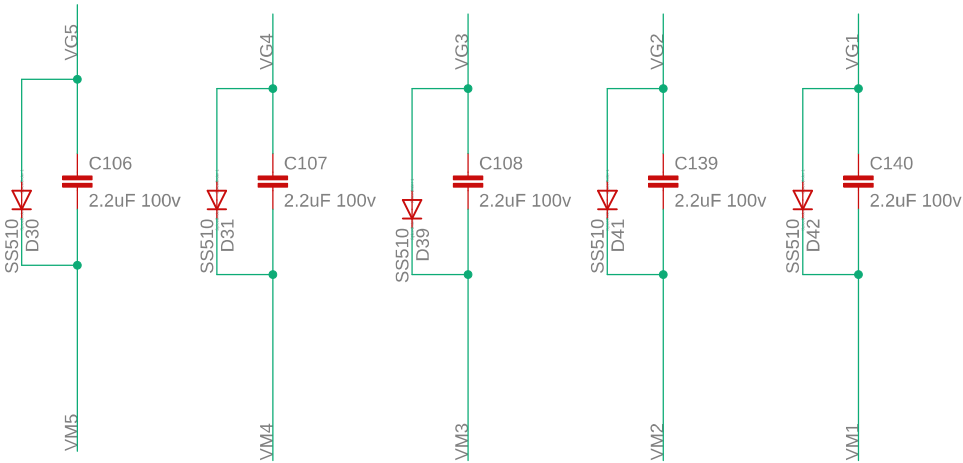
RS-485



eFUSE



PASS THROUGH POWER

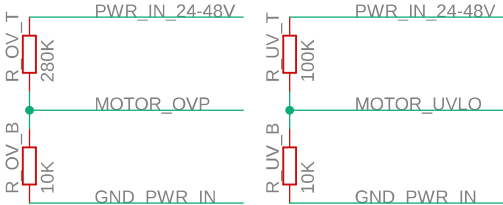


LM5060 configured for 24–48 V input range.

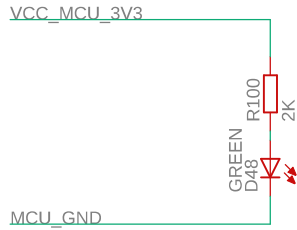
VDS current limit set to ~8 A using RS = 7.15 kΩ.

Calculation assumes BSC070N10NS3G MOSFET with ~11 mΩ hot RDS(on):
 $V_{DSTH} = R_S \times 16 \mu A - 25 \text{ mV}$.

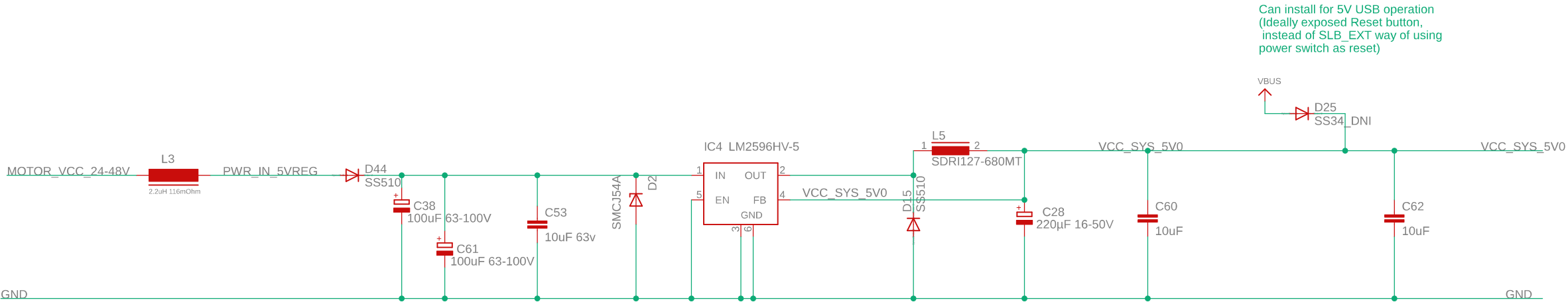
UVLO Turn-On 18.15 V
UVLO Turn-Off 15.51 V
OVP Shutdown (Trigger) 58.00 V
OVP Recovery 51.04 V



POWER LED

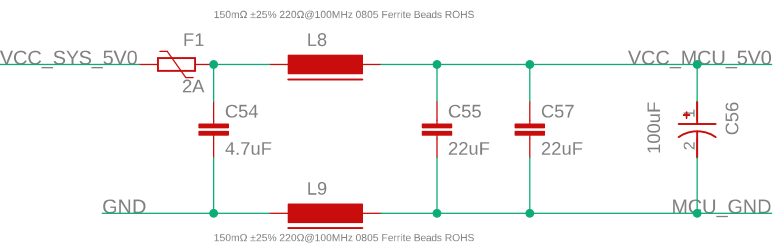


5V POWER SUPPLY

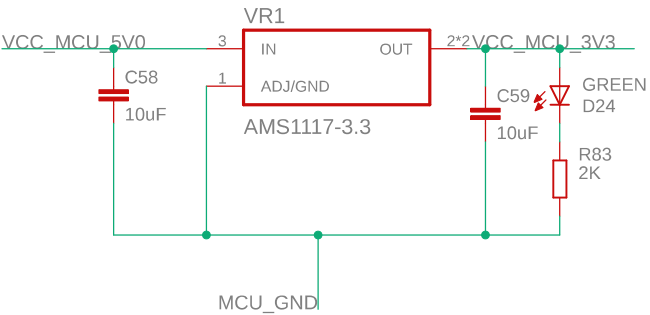


Can install for 5V USB operation
(Ideally exposed Reset button,
instead of SLB_EXT way of using
power switch as reset)

POWER FILTERING AND MCU RAIL



3.3V REGULATOR



EMERGENCY STOPS

